

Akash Gaur

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Summary

Builds end-to-end GenAI systems - from LangGraph agents and RAG pipelines to vLLM/Ollama production deployments - backed by FastAPI and vector database infrastructure. Delivered LLM-powered natural language querying for real-time analytics dashboards used in airport operations, reducing manual reporting effort for end users. Seeking a role where LLM engineering rigour and backend product quality need to coexist.

Experience

Software Developer — Kloudspot

Jun 2024 – Present

Promoted from intern to full-time based on performance (Jan–May 2024 internship).

KSpot AI Console – Local-First AI Agent Platform

- Built a self-hosted, Claude-Console compatible AI agent platform using **FastAPI**, **DeepAgents**, **PostgreSQL**, and **Docker**, enabling creation and management of production-grade AI agents and skills.
- Developed real-time **SSE streaming**, human-in-the-loop workflows, **multi-agent orchestration**, and a secure Docker-based **sandbox** for isolated execution of agent-generated code.
- Implemented provider-agnostic model integration (**AWS Bedrock**, **vLLM**, **Ollama**), encrypted MCP credential management, and persistent LangGraph state management with PostgreSQL.

LISA AI – Airport Operations Assistant

- Built an AI assistant for airport operations staff delivering real-time insights to **100+ daily users**, reducing manual query resolution time.
- Integrated **LangGraph** to orchestrate multi-step reasoning workflows with LLM agents, cutting average task completion steps by **~40%**.
- Implemented **text-to-MongoDB** and **text-to-SQL** query generation using fine-tuned Llama models, enabling non-technical staff to query live data in natural language.
- Utilized **Model Context Protocol (MCP)** for secure tool integration in an agentic AI framework.
- Generated dynamic **data visualisations (bar, line, pie charts)** from natural language queries, eliminating the need for manual BI report creation.

AI Workflow Automation using n8n

- Built **5+ automation pipelines** integrating AI services, APIs, and internal systems, saving an estimated **10+ hours/week** of manual processing.
- Developed event-driven workflows for data ingestion, transformation, and system triggers.

vLLM Inference Server Deployment

- Deployed **vLLM server** with OpenAI-compatible APIs for scalable LLM inference, supporting **concurrent multi-user** request handling.
- Optimised inference via **paged attention** and **GPU memory management**, achieving **~2× throughput improvement** over baseline.

Software Developer Intern — Kloudspot

Jan 2024 – May 2024

- Built a PoC for **building energy prediction** across commercial and residential properties, achieving **~85% model accuracy** on test data.
- Trained and compared ML models using **Scikit-learn** and **PyCaret** across 3 meter types (electricity, water, thermal).

Projects

PromptTokenizer - Visualize how LLM models tokenize your prompt

[Live Demo](#)

FastAPI · Python · tiktoken · Docker · Pytest · React · TypeScript · Tailwind CSS

- Built a **FastAPI microservice** supporting **49+ LLMs** with APIs for tokenization, model comparison, and pricing estimation.
- Designed a **pluggable adapter architecture** integrating multiple tokenizer engines and model registries.
- Optimized API through **startup preloading** and **lru_cache**-based singleton caching of tokenizers, model registries, and pricing services.
- Shipped with a **multi-stage Docker build**, Pytest suite, and a developer-focused React/shadcn UI.

- Built an **asynchronous AI chat backend** with FastAPI and MongoDB for persistent conversational memory.
- Integrated **8+ LLM providers** through a unified abstraction layer, enabling seamless model switching.
- Developed a **LangGraph agent workflow** for intelligent routing across chat, search, and reasoning pipelines.
- Implemented production-grade security with **JWT, bcrypt, RBAC, and email-OTP recovery**; shipped a pytest suite and containerized Docker/Gunicorn deployment.

Technical Skills

GenAI:	LangChain, LangGraph, LlamaIndex, DeepAgents, RAG, Prompt Engineering, n8n
LLM Ops:	vLLM, Ollama, Model Serving, LLM Deployment, GPU Inference
Backend:	FastAPI, Flask, REST APIs, JWT Authentication, Async APIs
Databases:	MySQL, MongoDB, Elasticsearch, Vector Databases
ML & Data:	Pandas, Scikit-learn, PyCaret, Keras, NLTK
Programming:	Python, C/C++, SQL
Visualization:	Matplotlib, Seaborn, Plotly
DevOps:	Linux, Docker, Git, GitHub, Bitbucket

Achievements

- **GATE Qualified — 2023** (Graduate Aptitude Test in Engineering, top national exam for CS/Engineering)
- Published a **Machine Learning research paper** on building energy prediction in MITS college journal (2023)
- Led a **team of 4 in a Microsoft-sponsored hackathon** at MITS (2022)

Education

Madhav Institute of Technology & Science, Gwalior B.Tech in Mathematics & Computing	2020 – 2024 CGPA: 9.45
Govt. School of Excellence No.1, Gwalior Senior Secondary (12th), MP Board	2019 – 2020 91.2%
Govt. School of Excellence No.1, Gwalior High School (10th), MP Board	2017 – 2018 93.2%

Certifications

Prompt Engineering for Developers — Udemy (2024)	[Certificate]
Machine Learning — SkillUp (2023)	[Certificate]
Data Science with Python — SkillUp (2023)	[Certificate]
Complete Python Programming — Udemy (2023)	[Certificate]